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Research Note
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Integrating the Development and Testing Life Cycle

The testing process must begin as early as possible and should be integrated into the AD life cycle in an iterative manner.

Core Topic
Applications Development: AD Support Technology

Key Issue
What strategies, technologies, tools and vendors will best enable the effective testing of client/server and other forms of cooperative-processing applications during the next five years?

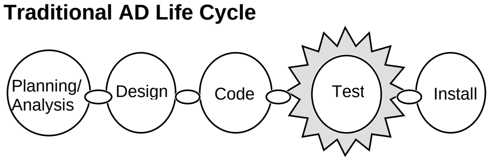
Strategic Planning Assumption
AD organizations that integrate the AD and testing life cycle through a unified methodology, and that begin testing activities in the planning/analysis phase, will experience a first-year, twofold increase in defect removal efficiency when compared to those that do not (0.7 probability).

For many organizations, application testing begins shortly after coding and right before installation. “Shortly” is often defined as the difference between the time that coding is thought to be complete and when the application is scheduled to be delivered — a period often susceptible to the whims of AD management and the de facto delays of most AD projects. Not only is the application “thrown over the wall” into quality assurance (QA) — if such a function exists — but often inadequate time is left for testing because budgeted testing time has been squeezed to meet application delivery dates (see Note 1). Unfortunately, this scenario often leads to poor software quality, subsequent costly application maintenance and unhappy business users.

Note 1
Defining QA
In this instance, we are equating QA with systems and acceptance testing. In many organizations, QA has more responsibility than just testing.

Industry testing pundits and leading vendors of automated testing tools have been promoting the notion of a “testing life cycle,” in which testing activities parallel AD activities. The basic premise is that the testing process should begin early in the AD life cycle, not just at the traditional testing phase (see Figure 1). There are several reasons to begin testing early. Test case design and test script construction can sometimes be as complex as the code it is intended to test. Consequently, ample time should be allocated to these efforts in order not to adversely affect the development schedule. In addition, test case design can potentially expose application design flaws earlier in the process, thereby lowering subsequent testing and defect removal costs.

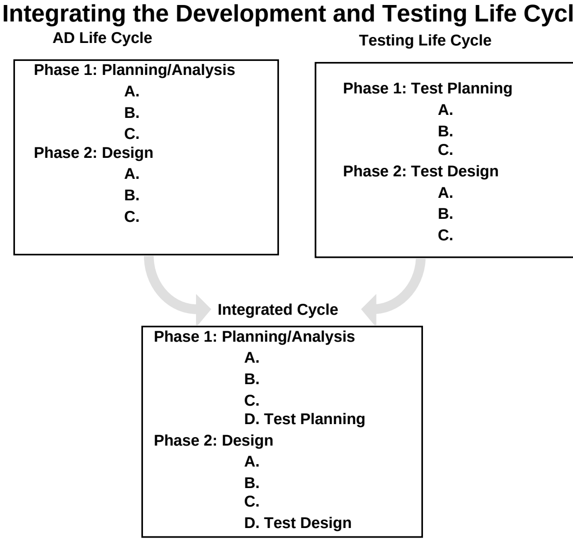
Figure 1
Traditional AD Life Cycle



Source: Gartner Group

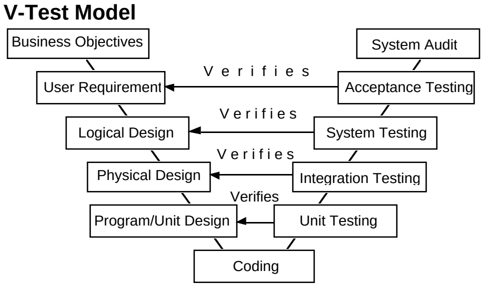
While we applaud and support the parallelism of AD and testing, we would prefer to extend it further in two ways. First, we believe that the testing process would be better served if it were to be formally integrated into the AD life cycle (see Figure 2). Parallel cycles, although synchronous, can potentially create the mistaken impression that AD and testing are separate processes. This is, in fact, not true. Optimally, there should be one development methodology that contains a software quality and testing thread in each and every phase. For example, during application requirements definition, certain application performance and service-level criteria should be specified in the test plan (e.g., the application must support up to 100 concurrent users). These criteria then feed test case development with a test requirement to load-test 100 or more concurrent users. Similarly, additional test cases are created during application analysis and design. During application code construction, reusable test scripts and test data are generated. Finally, during application testing, test scripts are executed and results are analyzed. This is one interpretation of what is sometimes referred to as the "V-test" principle or methodology, in which AD and testing activities eventually join at the "bottom of the V" during the traditional testing phase. Another interpretation depicts the V-test principle as stages of verification, with the coding effort residing at the bottom of the V (see Figure 3).

Figure 2
Integrating the Development and Testing Life Cycle



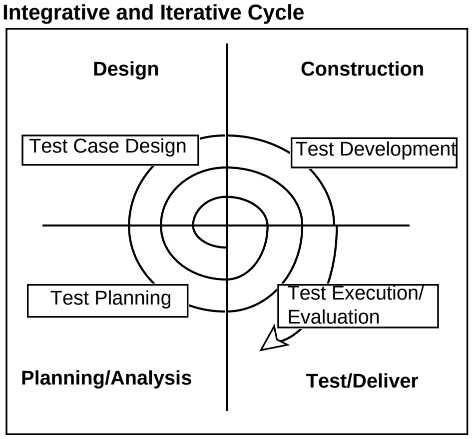
Source: Gartner Group

Figure 3
V-Test Model



Source: Gartner Group

Figure 4
Integrative and Iterative Cycle



Source: Gartner Group

Second, for C/S and RAD projects, it is imperative that the testing process become iterative in a fashion similar to the AD process (see Figure 4). In conducting such projects, successful AD organizations will drive through several iterations and exploit techniques such as JAD, prototyping and time-boxing to define and cement requirements, and to ensure timely delivery. Each iteration should produce a deliverable, and that deliverable (e.g., an application build) should be a target for testing.

Bottom Line: The testing process must begin as early as possible and should be integrated into the AD life cycle in an iterative manner. AD organizations that integrate the AD and testing life cycle through a unified methodology, and that begin testing activities in the planning/analysis phase, will experience a first-year, twofold increase in defect removal efficiency when compared to those that do not (0.7 probability).

Acronym Key

AD	Applications development
C/S	Client/server
JAD	Joint applications development
QA	Quality assurance
RAD	Rapid applications development